

Finding Regional Insights in BMI, Diabetes, and Hypertension

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Abstract

This project analyzes global health trends in BMI, Diabetes, and Hypertension between 1980 and 2014 using a Power BI Star Schema. By integrating population data with regional health metrics, the report shifts the focus from simple prevalence percentages to the absolute human impact. High-level findings reveal that while regions like Oceania face intensive clinical Red Zones with the highest percentages globally, Asia represents a massive-scale logistical crisis due to its vast population volume. The analysis also highlights the European Anomaly, where advanced medical interventions have successfully decoupled rising BMI from high blood pressure. These insights provide a strategic roadmap for tiered health investments based on regional needs.

Introduction

The objective of this project is to analyse the intersection of three critical global health indicators: Obesity (BMI), Diabetes, and Hypertension. As non-communicable diseases rise globally, this dashboard serves as a strategic tool for non-technical stakeholders to identify trends and regional health disparities. This report follows the **What, Why, and How** framework to justify the data abstraction and visualization choices made during the visualization process.

By using data from 200 countries, we aim to validate clinical claims such as the link between BMI and hypertension in diabetic patients on a global scale. A central challenge addressed in this study is the "statistical mask" created by looking only at percentages. By adding a population-weighted dimension, we move beyond health "velocity" to understand the true volume of suffering, ensuring that healthcare resources are directed where the human burden is most concentrated.

Data Overview

The analysis is based on three primary tabular datasets covering 200 countries.

1. BMI (Body Mass Index) Sheet

Representations: Tracks the prevalence of Obesity ($\text{BMI} \geq 30 \text{ kg/m}^2$).

Attributes:

Categorical (Nominal): Country/Region/World and Sex (Men/Women).

Temporal (Ordinal): Time series from 1975 to 2016.

Quantitative (Continuous): Prevalence value (percentage of population).

2. Diabetes Prevalence Sheet

Representations: Tracks age-standardized diabetes prevalence.

Key Feature: Data is age-standardized to ensure fair comparison between countries with different demographic age profiles (e.g., comparing an aging population to a younger one).

Attributes: Categorical and Quantitative attributes matching the BMI sheet, spanning 1980 to 2014.

3. Raised Blood Pressure Sheet

Representations: Tracks the prevalence of Hypertension.

Key Feature: While BMI and Diabetes often trend upward together, Blood Pressure may show different patterns in developed regions due to medical intervention, providing a critical "novelty" angle for analysis.

Attributes: Categorical and Quantitative attributes spanning 1975 to 2015.

Data Preparation

To ensure a realistic and credible solution, significant modifications were performed at the data source level.

1. Hierarchical Mapping (The Region Table)

Modification: Integrated a custom mapping table based on the FAO/Impact Global Health standards.

Justification: This adds a Region > Country hierarchy. It prevents "Visual Noise" by allowing high-level regional comparisons while maintaining the ability to drill down into specific national data.

2. Temporal Filtering

Modification: Filtered all datasets to a common range of 1980–2014 unless comparing within a single sheet, then all of their dates/years will be used.

Justification: Aligning the timeframes ensures "Quality of Findings" and prevents temporal bias when calculating correlations between the three metrics.

3. Added Population Table:

Added Population table region wise and year wise so we can see how population plays a key role and how much the data stories are changed once we see it population wise. Mapped it to the region and Year table to build a connection with so it can be filtered with every other health sheet.

This will enable us to view the impact and scale of every region and give better insights.

A significant challenge in the data modelling phase was the granularity mismatch between the health datasets (recorded at country level) and the population data (recorded at region level). To maintain mathematical integrity and avoid inflating affected-population counts, the analysis required aggregating prevalence to the regional grain before applying population weights.

4. Modelling: The Star Schema

Structure: Created Dimension Tables (Geography, Year, Sex, Population) to act as a central hub for the three Fact Tables.

This structure enables a single Year and Region slicer to control all visualizations simultaneously, improving dashboard "Engagement" and "Accessibility".

Analysis Questions

Clinical Correlation & Global Validation: To what extent does the established link between BMI and hypertension in diabetic patients as evidenced by Tseng (2007) apply universally across all global regions?

Regional Disparity & Growth Velocity: Which regions demonstrate the highest velocity of obesity growth, and which nations remain statistically significant low-prevalence outliers?

Metabolic Interaction & Gender Shifts: How does the correlation between high BMI and diabetes prevalence shift across different regions, and why do men in certain areas overtake women in diabetes rates despite lower obesity?

Trend Divergence & Medical Intervention: How do longitudinal trends in hypertension deviate from rising BMI trajectories, and what does this "European Anomaly" suggest about the effectiveness of healthcare intervention?

Scale of Impact: How does the integration of population data shift the narrative from health "velocity" (percentage of change) to health "volume" (absolute human burden)?

Visual Encoding and Dashboard Interactive Features

1. Visual Encoding and Mapping

Line Charts: Used for temporal trends as Position on a common scale is the most effective channel for showing change over time.

Scatter Plots: Implemented to visualize the correlation between BMI (X-axis) and Diabetes (Y-axis).

Color Channels: Used to distinguish between regions to identify region-based Health metrics.

2. Interactive Features

Cross-Filtering: Selecting a specific region on a map filters the trend lines for all three-health metrics.

Findings

Let's look at the first analytical question that I have mentioned. Where I hypothesize that the evidence in the article showed that BMI and hypertension are connected to diabetes I would like to hypothesize that this would be true for all the people in different countries / region as the symptoms of diseases are similar in most cases. To see if this is true or how much these three correlate with each other I will visualize the data across similar Years (1980-2014) and take the average of each of the continuous variables over multiple countries under each region and see if there is a difference in pattern when we dive even deeper as some places have different practices than other countries depending on the country within that region as well and since we have 200 countries comparing each country with each other will not be visually understandable however we can compare the top 3 best and worst countries for each region from the 200 countries as well which will come later in the analysis.

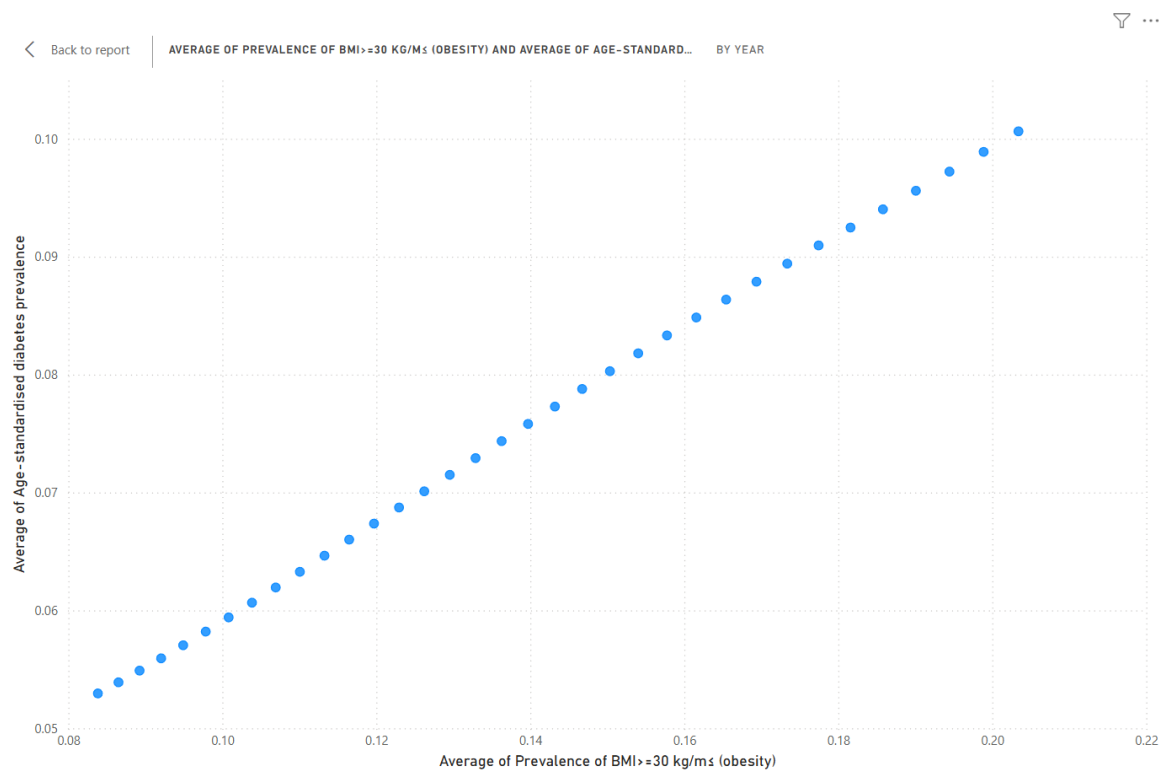


Figure 1. Scatter Chart of Average Prevalence of Obesity vs Average Prevalence of Diabetes (1980 to 2014)

First let's look at this scatter chart which shows the average Obesity percentage of all 200 countries over each year and similarly average of diabetes patients percentage of all

countries over each year. At first let's look at BMI which shows an increasing line which means that over the years the percentage of Obesity increased for a majority (if not all) of countries as the average increased overtime. And Similarly for the Diabetes line it shows steady increase over time as well which does show that increase in diabetes can be related to increase in obesity which also supports the research of the evidence-based article Samuel et al. (2022) where Diabetes patients have high prevalence of BMI. Which means Obesity has a high relation in causing diabetes.

Furthermore another research article says in its conclusion, "BMI/Obesity is significantly linked to blood pressure/hypertension throughout the range of BMI in diabetic patients in either sex regardless of a previous hypertension history.", this was true for Taiwanese patients as the all the analysis conducted was using Taiwanese patients data. Now using this theory, we can hypothesize that this should be true for every country/region and to see if there is any relation between diabetes, high blood pressure and Obesity by visualizing the data we have available.

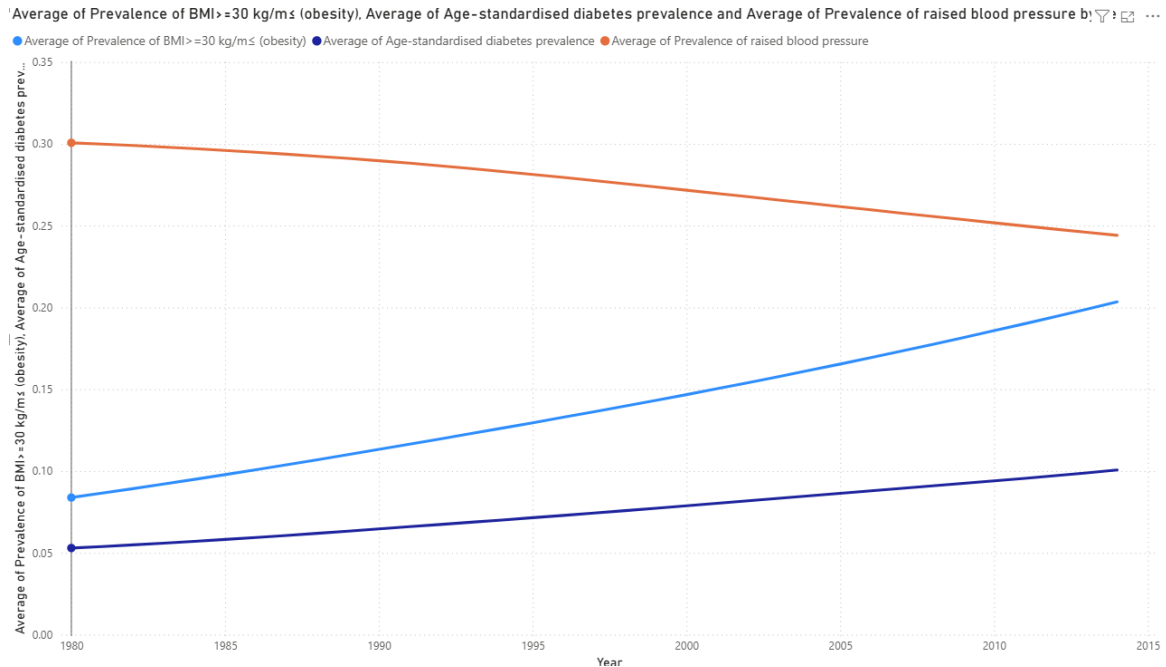


Figure 1.1. Average Prevalence of Obesity vs Average Prevalence of Diabetes vs Average Prevalence of High blood pressure (1980 to 2014)

Now when we look at the above visualization after adding the prevalence of high blood pressure alongside the visual of BMI and Diabetes, we can notice that the probability of having high blood pressure in individuals for all countries decreased over time. Now with this there are multiple things that can be deduced where a deeper understanding might be required to get a better understanding of the given data. First simple deduction that can be made is that according to the research article mentioned before this shows that probability of high blood pressure might not be related to Obesity or Diabetes but since this data includes avg of every country it might not be as accurate and to see a decrease in probability over the years can also mean that the evolution of healthcare over time could have posed a

key role in controlling high blood pressure hence the decrease over time but this might not be true for all countries/regions as level of advanced medicine varies for each country/region. So, to have a deeper understanding of this chart we will visualize this chart country wise and see what difference we see when we visualize this region wise.

A General Regional Overview of the Data:

Prevalence of Obesity Chart:

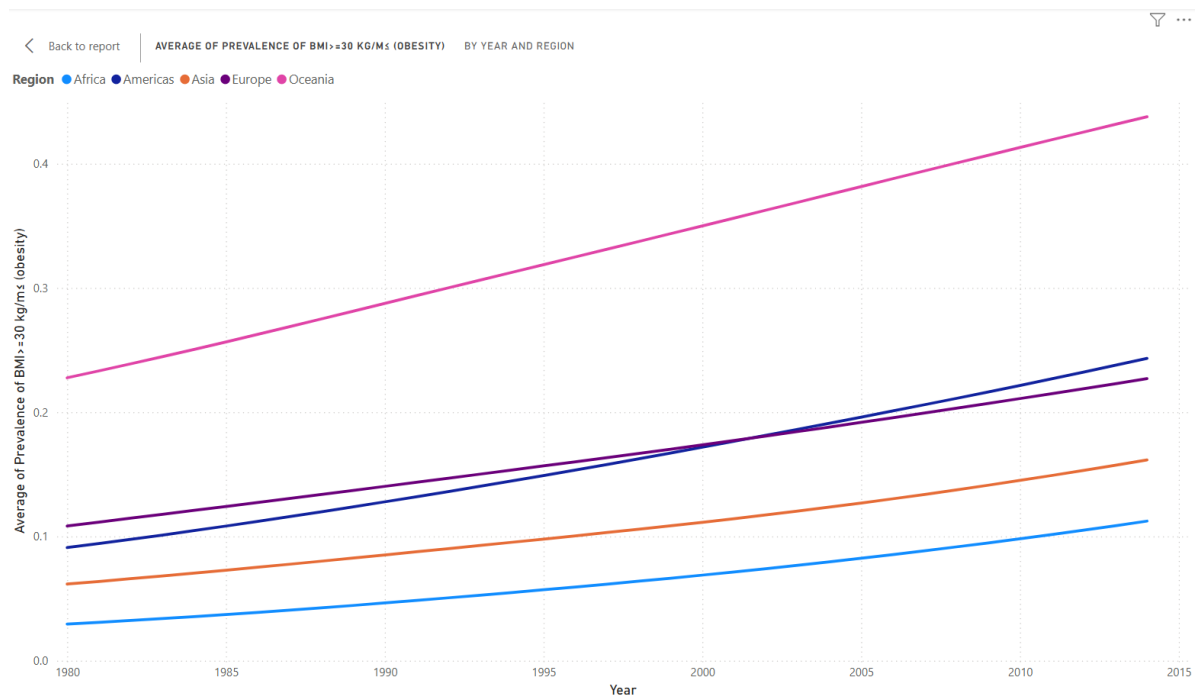


Figure 2. Region wise Average Prevalence of Obesity (1980 to 2014)

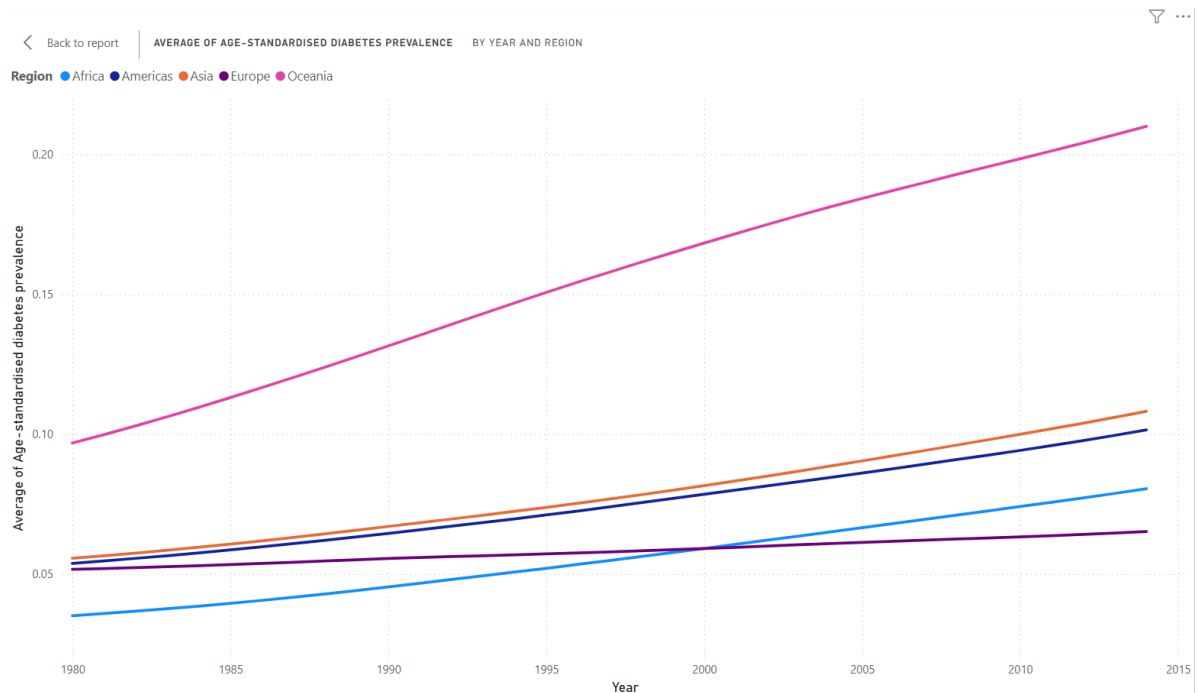


Figure 2.1. Region wise Average Prevalence of Diabetes (1980 to 2014)

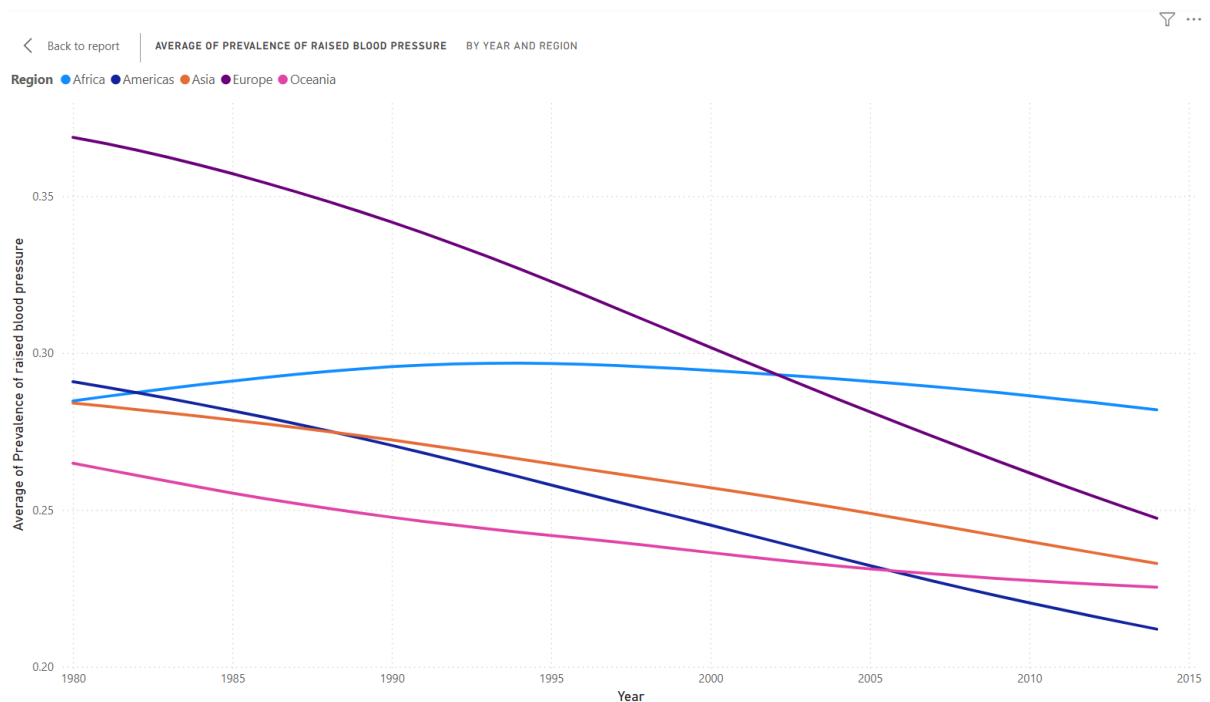


Figure 2.2. Region wise Average Prevalence of High Blood Pressure (1980 to 2014)

Across all regions, there is a steady and visible increase in obesity prevalence over the 34-year period. A standout finding in this dataset is the prevalence in Oceania, which reached 0.44 by 2014 nearly double that of the Americas, the second-highest region at 0.24.

This massive gap represents a significant "anomaly" in the global data. From a critical perspective, such a high average for an entire region suggests that the data may be driven

by specific high-intensity clusters. This finding serves as the primary motivation for implementing a Sub-region hierarchy in the dashboard; without drilling down into areas like Polynesia or Micronesia, the high regional average might unfairly misrepresent the trends in Australia and New Zealand.

Meanwhile, Africa and Asia maintain the lowest prevalence rates at 0.11 and 0.16 respectively, though their upward trajectories confirm that obesity is a universal challenge regardless of the starting point.

There is a clear pattern, as Obesity increases, Diabetes prevalence follows in almost every region. Oceania leads both categories, reaching 0.21 for Diabetes by 2014. However, Europe is a major exception. Even though Europe's obesity is close to the Americas (0.23 vs 0.24), it has the lowest prevalence of Diabetes and the smallest change over 34 years (only 0.03). This is a "critical finding" because it suggests that Europe might have better early intervention or that the age-standardized nature of the data shows a different health reality there.

For High Blood pressure, across all regions, there is a steady and visible decrease in high blood pressure prevalence over the 34-year period. A standout finding in this chart is that Europe has the highest decrease in prevalence of high blood pressure than any other region (dropping from 0.38 in 1980 to 0.24 in 2014) which must mean that Europe must have invested a lot in controlling high blood pressure in every country because of which the average drop became so steep.

While every Region seems to show a steady decrease in the prevalence of High Blood pressure Africa seems to be unable to control the rate of high blood pressure and because of this along with rising obesity might cause Africa to have increased prevalence of Diabetes.

While BMI and Diabetes move together, Raised Blood Pressure shows a different story in some places. In the Europe and Central Asia region, BMI is rising, but Blood Pressure is actually going down or staying flat. This is a very interesting finding. It suggests that even if people are getting heavier, better healthcare and medicine are helping to control their blood pressure, which might prevent heart issues even if they still have a risk for diabetes.

Female vs Male analysis:

For this analysis I made a custom measure for each of the continuous variables where I subtract the Male values from Female values, and the result will tell the following:

If positive: Female prevalence is more than male

If negative: Male prevalence is more than female

If 0: both have the same prevalence

The Formula: Gender Gap = [Female Prevalence] - [Male Prevalence]

I named the new measures as following for better understanding in the visualizations:

Obesity Prevalence Gap (Female vs. Male)

Diabetes Prevalence Gap (Female vs. Male)

Blood Pressure Gap (Female vs. Male)

Now to study male vs Female we will need to go by each region to have a better understanding as it can vary from region to region and even sub regions and with that information, we can figure out which region, and which gender might need more attention/investment in healthcare. First let's see a general overview of Male vs Female Region wise for each sheet.

I used a Clustered Bar Chart for the Gender Gap because placing the bars for Men and Women side-by-side seemed like the most effective way to compare two categories accurately.

Prevalence of Obesity (1975 – 2016)

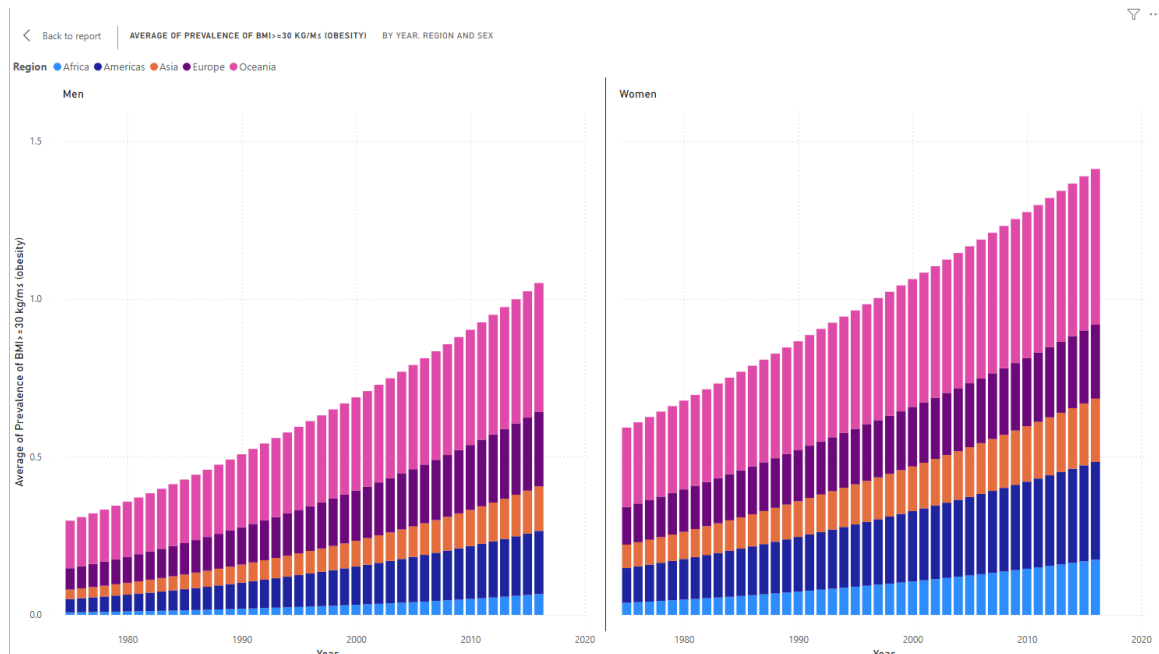


Figure 3. Region wise Average Prevalence of Obesity (Men & Women) (1975 to 2016)

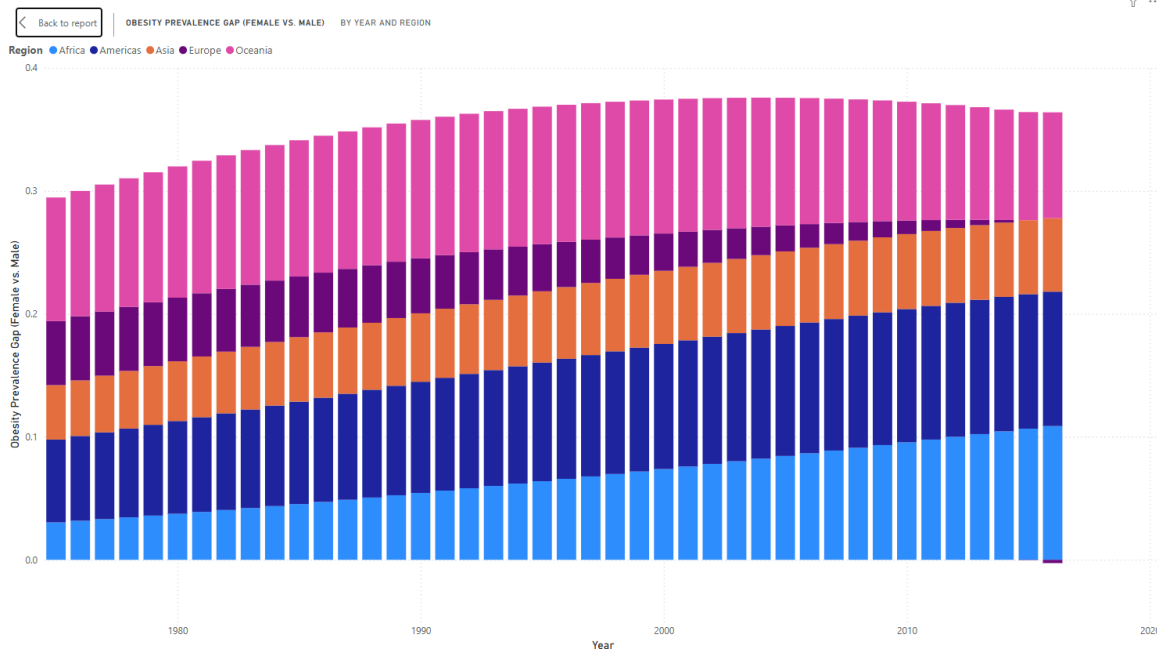


Figure 3.1. Region wise Obesity Prevalence Gender Gap (Women vs Men) (1975 to 2016)

First let's look at Figure 3, we can notice that prevalence of Obesity has an increasing trend in Men and women. Having a closer look we can notice that women seem to have a higher number of prevalence in obesity than men in all of the regions and this is where the gender gap measure will help us in having a closer look as to how much more women and men are divided when it comes to obesity which is in Figure 3.1.

First look at chart 2 tells us that prevalence of obesity in women in Asia, Africa, Americas and Oceania are consistently increasing (even if it is a very slight increase) throughout the year with the sole exception of Europe where that trend decreased and even reached 0 and even went slightly negative which shows they managed to change or help control obesity to the point where the obesity rate is equal to men which also shows that they might have focused in womens healthcare to have some control over the increase while the other regions are struggling to control prevalence of obesity in women.

When looking at obesity, there is a consistent pattern across almost every region, women have a higher prevalence than men. Americas, Oceania, Asia, Africa. In these regions, the gap is very noticeable, with women consistently staying ahead of men in obesity rates throughout the entire 34-year period.

This finding suggests that health actions for obesity might need to be specifically targeted toward women's lifestyle and nutritional needs in regions like the Americas, Oceania, and even Africa. Though in Europe we do notice that men have a slightly higher increase at the end of 2014 since it passed women's threshold hence Europe might have been lacking in promoting healthcare for men.

Prevalence of High Blood Pressure (1975 – 2015)

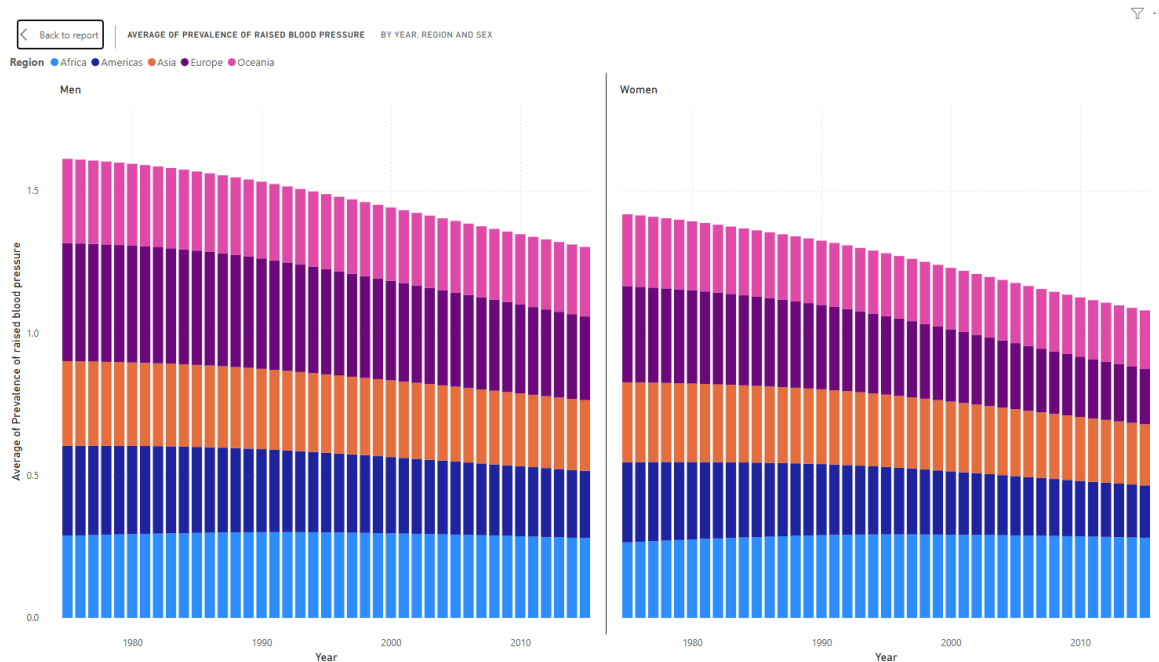


Figure 3.2. Region wise Average Prevalence of High Blood Pressure (Men & Women) (1975 to 2015)

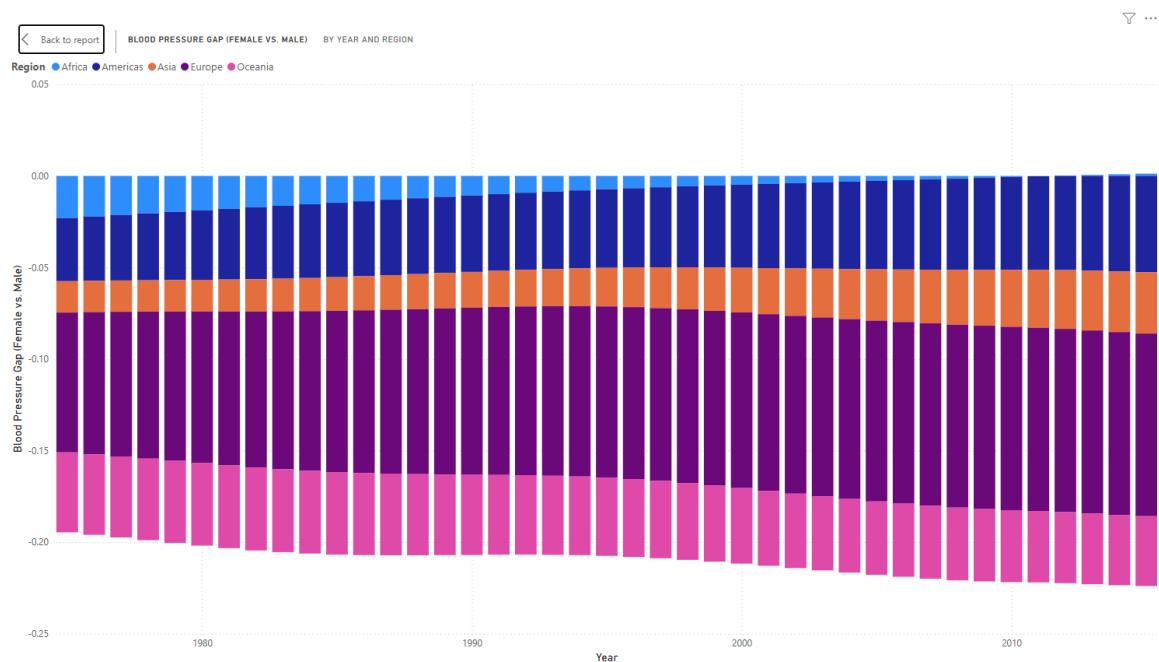


Figure 3.3. Region wise High Blood Pressure Prevalence Gender Gap (Women vs Men) (1975 to 2015)

While obesity is generally higher in women, the data for Raised Blood Pressure shows a that high blood pressure affectees are more men than women.

In Africa & Asia, the gender gap for hypertension is quite narrow. In Africa, the prevalence is almost equal for both sexes even though it's slightly negative, remaining mostly consistent

over the 34-year period. In Asia, we see a similar pattern where the difference is minimal, suggesting that environmental or lifestyle factors might be affecting both genders equally.

In Europe while obesity is rising, blood pressure trends for both men and women are actually stabilizing even though majority of the population affected by high blood pressure are men as shown in Chart 2 they are still stabilizing over the years. This supports the idea of "medical intervention" mentioned earlier in the report.

In Oceania & Americas, men tend to have slightly higher rates of raised blood pressure than women.

This finding suggests that in terms of high blood pressure every region through out the years have found ways to minimize the risks of high blood pressure and they continue to decrease as the healthcare, awareness and medicines preventing high blood pressure becomes more accessible and the risk seems to be decreasing.

Prevalence of Diabetes (1980-2014)

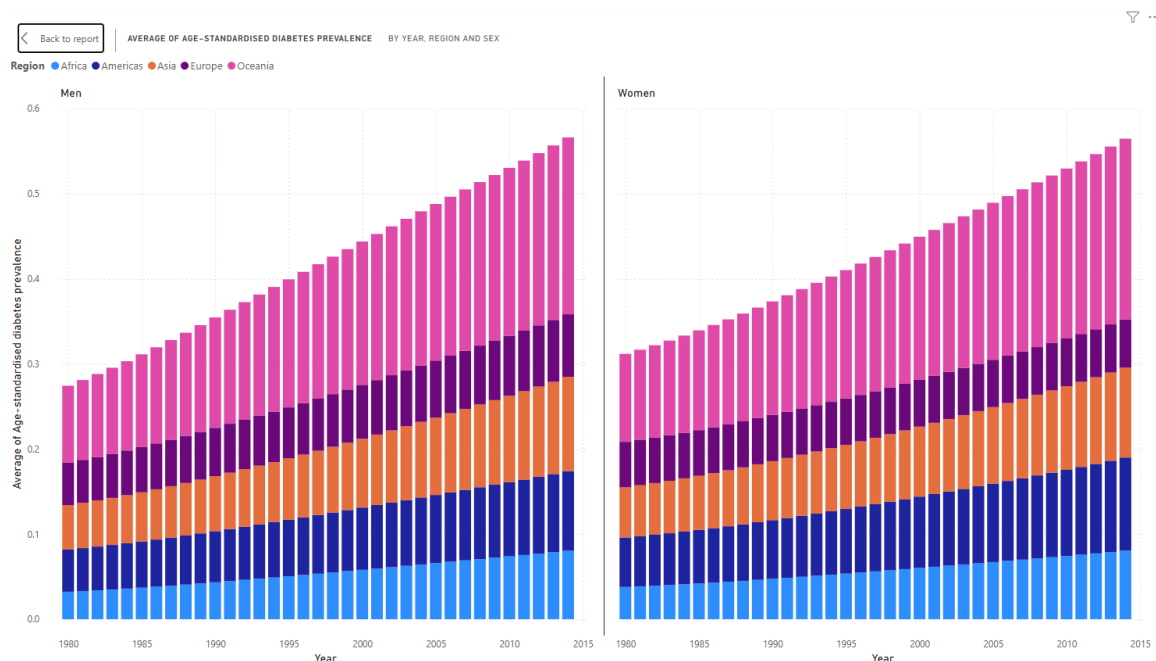


Figure 3.4 Region wise Average Prevalence of Diabetes (Men & Women) (1980 to 2014)

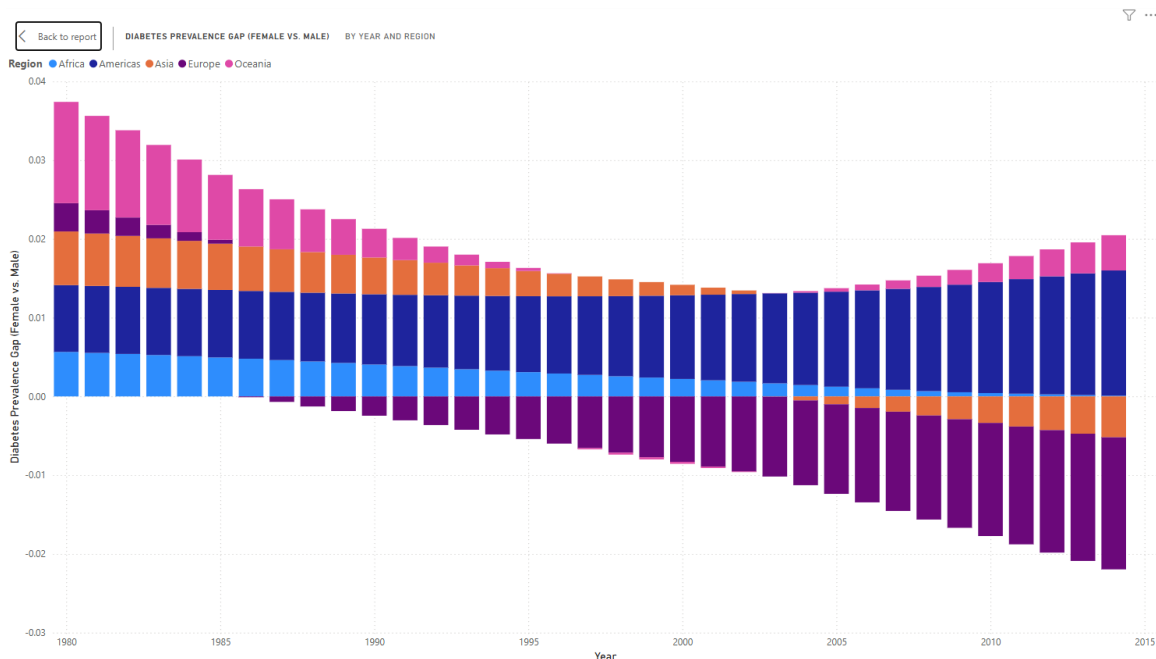


Figure 3.5. Region wise Diabetes Prevalence Gender Gap (Women vs Men) (1980 to 2014)

The most surprising discovery is that diabetes trends do not always follow obesity trends when it comes to gender.

In Asia & Europe in the early years (the 1980s), women had higher diabetes rates. However, a major shift occurred after 1986 in Europe and 2003 in Asia where men overtook women. Men now have a higher prevalence of diabetes in these regions even though women often have higher obesity.

Americas region is different. Women have always had a higher prevalence of diabetes, and that gap has even grown slightly over time.

In Oceania the gap was wide at first, then balanced out, and shifted back toward women after 2003.

This shift in Europe and Asia is a critical finding. It shows that even if women are more likely to be obese, men in these regions might be more biologically susceptible to diabetes or have different lifestyle risks that lead to the disease faster.

Population-Weighted Impact Analysis:

Now after looking at the data we can notice how much increase or decrease each region has for each prevalence altogether but in terms of the scale and impact we can not figure that out unless we have a better understanding of how population can change the scale of even the tiniest of percentage. Hence I made a formula to see what percentage of the population is affected which will give us a better understanding of how many people are actually affected so even if the rate of change might be low for a region but if their population is a lot compared to other regions then that will show a completely different story. So now let's see for each variable the difference and the impact keeping the previous results in mind.

First lets see a general breakdown of population Regionwise and Year wise over the years (1980 to 2014):

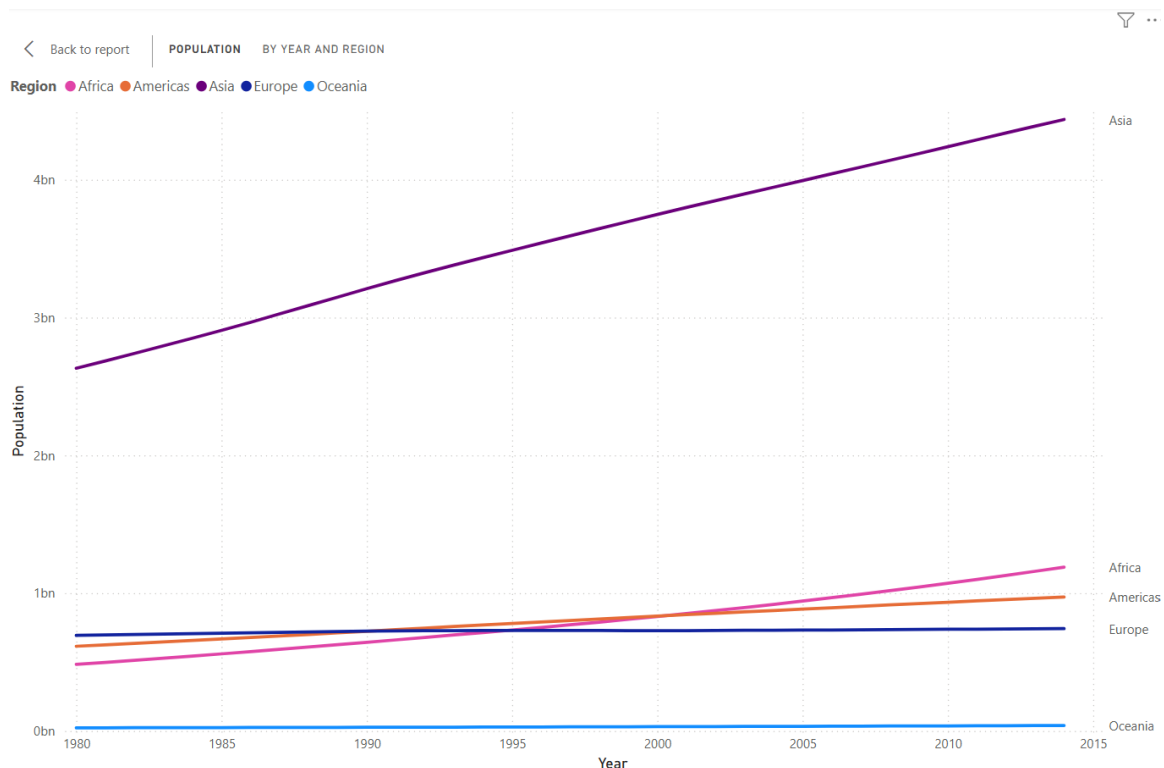


Figure 4. Region-wise Population (1980 to 2014)

A critical finding in the demographic narrative is the stark contrast between health velocity (the rate of change) and health volume (the total number of lives affected). While the previous sections focused on percentages, incorporating population data reveals a massive disparity in global health responsibility.

The data confirms that Asia maintains the highest absolute population of any region, coupled with the most aggressive rate of demographic increase. This creates a significant "statistical mask" in health policy.

When viewed solely by prevalence rates, Asia's diabetes and obesity values appear relatively low (e.g., 0.08 for diabetes).

However, when the Scale of Impact is calculated, Asia emerges as the global epicenter of the metabolic crisis. Because the population base is so vast, even a low percentage represents a higher total number of affected individuals than in high-prevalence regions like Oceania.

To account for this, the dashboard utilizes a Weighted Population Metric. This provides a clearer picture of the actual human burden, proving that the most urgent public health target is not necessarily where the percentage is highest, but where the volume of suffering is most concentrated.

Europe and Oceania remained stable throughout the 34-year period. Oceania's stable population means that its extreme health rates (the highest in the world) are concentrated within a smaller, more localized group. This makes the region a high-intensity target for clinical intervention.

Africa and The Americas occupy the middle ground, showing a steady but "controlled" increase in population.

Africa's steady population growth, combined with its rising BMI trends, creates a time-bomb scenario. As the population expands, the absolute number of diabetes cases will grow exponentially even if the prevalence rate remains steady.

In the Americas, the population increase is closely synchronized with the rise in obesity, indicating that health infrastructure must expand at the same rate as demographic growth to prevent a total system failure.

Regional Impact of Obesity:

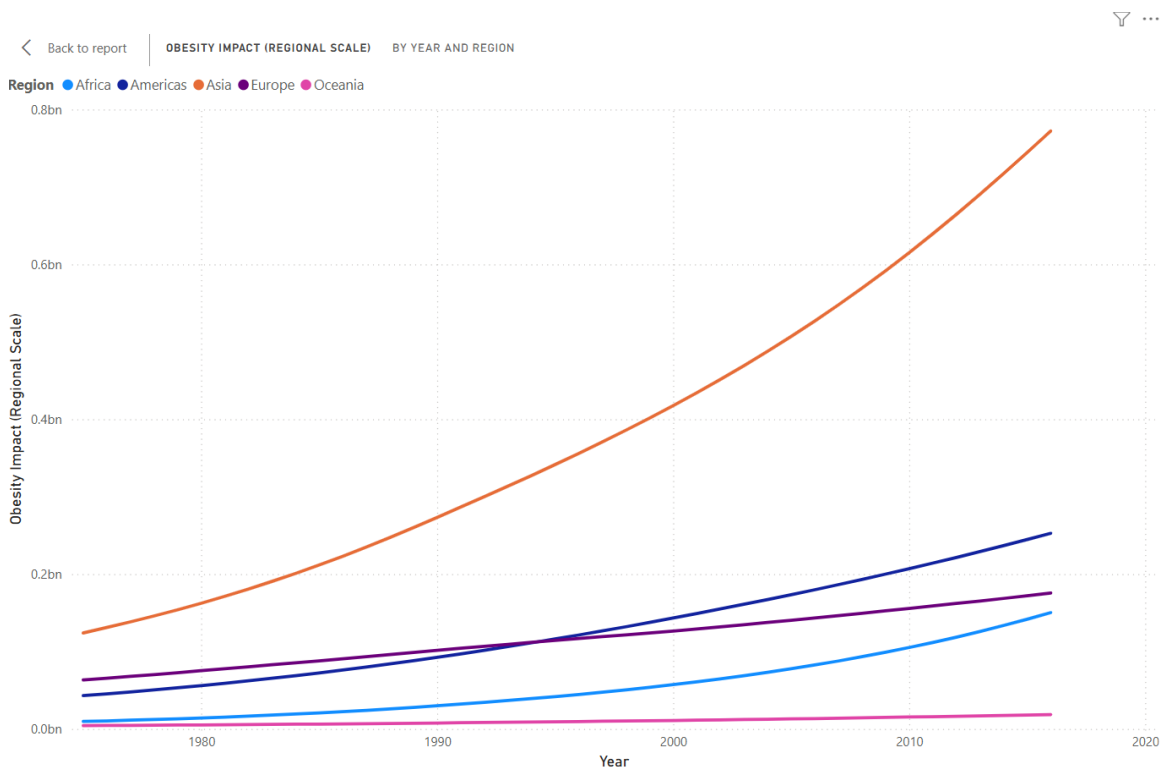


Figure 4.1. Regional Impact of Obesity (1980 to 2014)

Looking at the chart, Oceania remains the most alarming narrative. The average prevalence of obesity here is 0.44 by 2014 (viewing previous obesity chart) which is so extreme that it sits nearly double that of the next highest regions even though it is a straight line since Oceania's population is controlled but this still signifies that 44% of their population have prevalence in obesity.

Oceania here represents the Extreme Case. Despite having a smaller population, the intensity of the crisis is a lot. This identifies Oceania as a Red Zone, where significant and necessary steps must be taken in healthcare in order to lower the obesity rates which apart from diabetes can still cause multiple other health risks.

The Americas having 0.24 by 2014 and Europe 0.23 by 2014 are almost identical in their obesity trends. Their trend does not seem as excessive as the trend shown for Asia and their population increase

Asia having a relatively low prevalence of 0.16 by 2014, in this chart the slope of that line is critical and more exponential. While Asia's obesity percentage is lower than the Americas or Europe, its population scale turns any increase in obesity into a much larger population affected by obesity. This confirms that Asia is following the Western trend but at a much larger scale. As Asia's population continues to rise, even small increases in obesity prevalence will result in hundreds of millions of people entering the "high-risk" category for diabetes and heart disease.

Africa's line shows the most significant change in terms of relative growth after Asia. Starting at a very low baseline of 0.02, Africa has climbed to 0.12. While 12% sounds low, it represents a high increase in prevalence over 34 years which also means there will be a higher risk in the prevalence of diabetes since Obesity is related to Diabetes.

This finding suggests that Asia, America and Africa need to focus more on combatting high prevalence of obesity and they also need to focus more on women as they are the ones most affected in these regions and this might help with preventing prevalence of diabetes as well.

Regional Impact of Diabetes:

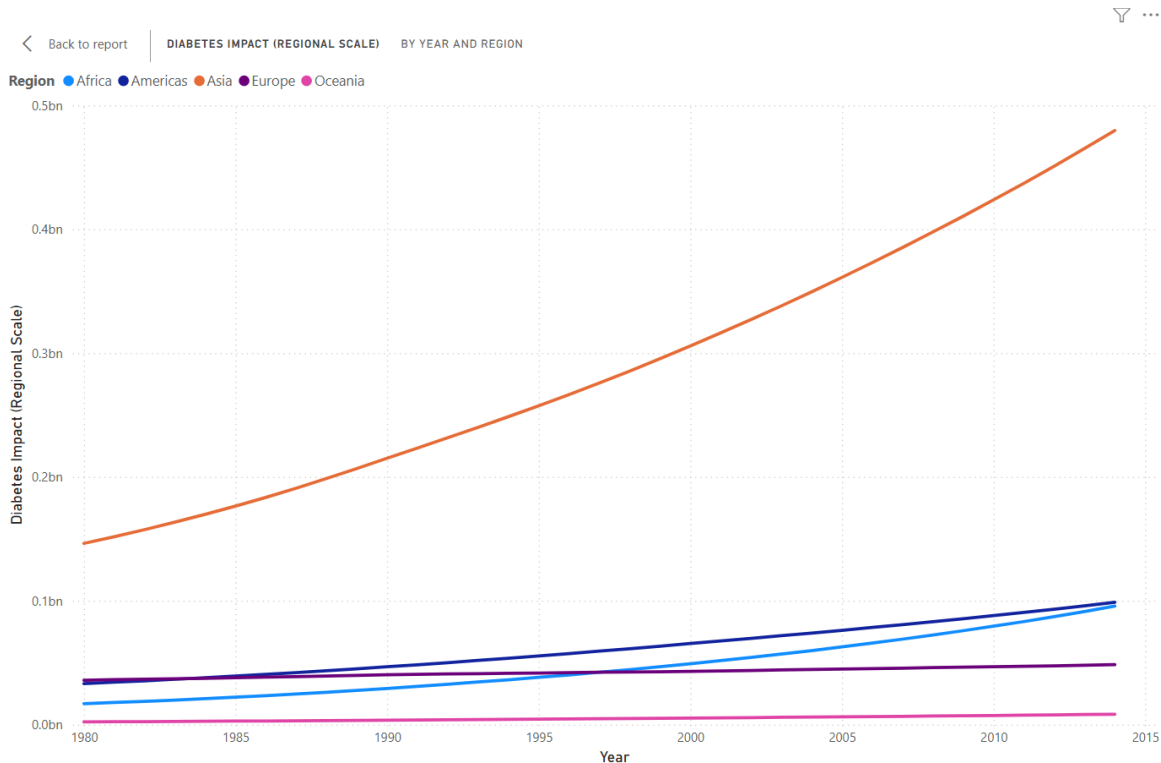


Figure 4.1. Regional Impact of Diabetes (1980 to 2014)

While the earlier analysis identified Oceania as the region with the highest percentage of diabetes 0.21, this visualization reveals a different reality when accounting for demographic weight.

Even though Asia exhibits a significantly lower diabetes prevalence 0.08 compared to Oceania, the absolute number of individuals affected is dramatically higher in Asia. This is driven by two primary demographic factors identified in the data:

Baseline Volume: Asia's massive total population acts as a multiplier for every percentage point of disease.

Growth Velocity: The rate of population increase in Asia is far more aggressive than in any other region.

This finding proves that while Oceania faces an intensive clinical crisis, Asia faces a more complex problem. A 1% increase in diabetes in Asia represents millions more patients than a 10% increase in Oceania. This discovery fundamentally shifts the story of the report: we can no longer define "highest risk" solely by percentage; we must define it by the total human burden on the healthcare system.

Africa's diabetes prevalence 0.08 is equal to Asia's, but the rate of the population growth is what matters here. Because Africa's population is rising so steadily, the total number of people affected is increasing every single year even while the percentage stays stable. This

creates a "long-term burden" that will eventually require more medical resources than Europe or the Americas if the trend continues.

Europe & Oceania regions have a very different story because their populations are stable.

Even though Europe has the lowest prevalence 0.07, the stable population means the "total burden" isn't exploding like it is in Asia. This allows European healthcare systems to plan more effectively because the "volume" of new patients is predictable.

Oceania is the most extreme case of the "Prevalence Paradox." Oceania has the highest percentage 0.21 by 2014, but because the population is so small and stable, the total number of people affected is the lowest of all regions.

The Conclusion: Oceania is a "High-Intensity" crisis (a lot of sick people in a small space), while Asia is a "Massive-Scale" crisis (a lot of sick people spread across a continent).

High Blood Pressure:

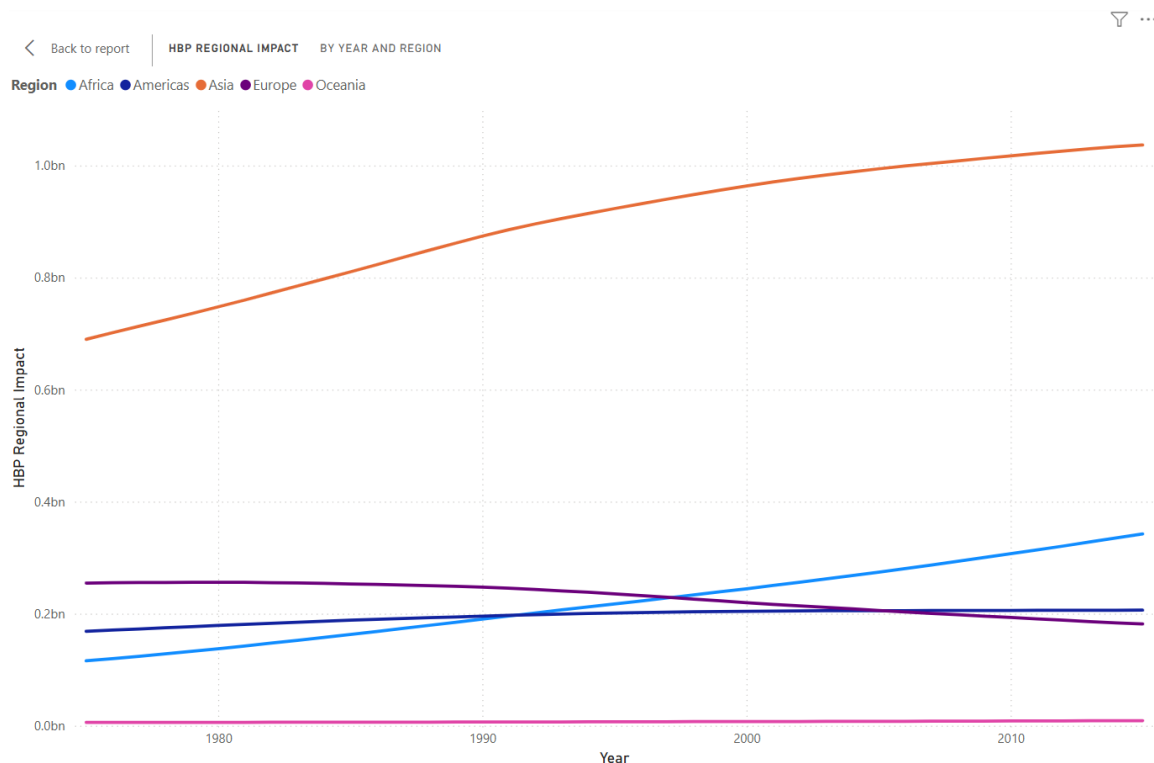


Figure 4.2. Regional Impact of High Blood Pressure (1980 to 2014)

When we transition from viewing High Blood Pressure (HBP) as a mere "probability" to an "Absolute Human Impact" metric, the global narrative shifts from one of stability to one of increasing urgency.

In the earlier prevalence-only charts, Africa appeared to have a very flat and stable hypertension rate, remaining consistent at approximately 0.28.

While the probability (percentage) of an individual having hypertension in Africa has not changed significantly since 1980, this chart shows that the Total Affected Impact is rising.

Because Africa's population is among the fastest-growing in the world, a "stable" percentage is actually a masking effect. Every year that the rate stays at 28%, millions of new individuals are entering the healthcare system with high blood pressure. This proves that for African health ministers, the crisis is expanding at the speed of their population growth, even if the individual risk remains unchanged.

As identified in our gender analysis, HBP is a male-dominant trend. When combined with Asia's massive demographic footprint, the "Impact Scale" identifies this as the largest single cardiovascular risk group in your dataset.

Asia may not lead in prevalence (percentage), but it dramatically leads in Human Volume.

The "Expected Value" (Probability x Population) for Asia creates a logistical challenge that no other region faces. This justifies a recommendation for scalable, low-cost medication distribution and community-led screenings. In Asia, the battle against hypertension is a battle of numbers; individualized clinical care cannot keep pace with the sheer volume of patients identified by your chart. Another finding for Asia is that even though the rate of high blood pressure seemed to be decreasing the increase in population still showed that more and more people are still going to be affected even if the rate is low just because the increase in population is very high.

Europe shows a very interesting find. Even though their population increased their impact actually decreased which shows that the total Affected Impact is stabilizing or even slightly decreasing in certain sub-regions.

This confirms the "European Anomaly" discussed earlier. Europe is successfully decoupling BMI from Blood Pressure. By lowering the probability (prevalence) through medical intervention (antihypertensives and salt-reduction policies) faster than the population can grow, Europe is successfully managing the "Human Burden."

Another Finding in this chart shows us that High Blood Pressure is in-fact high and going up alongside Obesity and they are actually correlated each other and hence correlated with Diabetes as well since if we look at Asia in all 3 charts all of the lines are going up the only difference is the increase in high blood pressure is decreasing but the volume is still increasing as the population increases.

Strategic Solutions:

Addressing the Scale of Impact

Solution: Tiered Infrastructural Investment (Target: Asia)

The Problem: The sheer volume of patients in Asia means that traditional, one-on-one doctor visits cannot scale to meet the demand.

The Data-Driven Solution: Investment must be directed toward Digital Health Infrastructure and automated screening tools. By utilizing AI-driven diagnostics that can process thousands of patients simultaneously, Asian healthcare systems can manage the "Scale of Impact" identified in our population-weighted charts.

Expected Outcome: A reduction in the wait-times for diagnosis, preventing the "population surge" from overwhelming existing hospitals.

Solution: Intensive Clinical Intervention (Target: Oceania)

The Problem: While the population is smaller and stable, the prevalence is so extreme (0.21) that it indicates a localized epidemic.

The Data-Driven Solution: Implementation of High-Intensity Clinical Taskforces. Because the population is "controlled" and stable, the solution should focus on door-to-door screening and aggressive nutritional mandates to "break" the high-prevalence trend.

Expected Outcome: A flattening of the prevalence curve in a region where the disease has become culturally and environmentally embedded.

General Solution:

Apart from this some solutions can be gathered by further researching every region and their countries and finding out which country among each regions have the lowest rates and what are those countries doing different than the ones that have high prevalence rates and compare their actions taken against all these health issues and learn from them and this is very easy to figure out which countries we are looking for. For example if we take Asia we can easily figure out which countries need immediate attention and which countries we can learn from or figure out why they have low prevalence rates.

Top 5		Bottom 5	
Country	Average of Prevalence of BMI >=30 kg/m ² (obesity)	Country	Average of Prevalence of BMI >=30 kg/m ² (obesity)
Jordan	0.25	Viet Nam	0.01
United Arab Emirates	0.25	Bangladesh	0.01
Saudi Arabia	0.25	Timor-Leste	0.01
Qatar	0.28	Cambodia	0.01
Kuwait	0.31	Nepal	0.02
Total	0.27	Total	0.01

Figure 5. Top 5 and Bottom 5 Countries with Average Prevalence of Obesity (1980 to 2014)

In this example I showed a total average over the 34 years for each country and got top 5 and bottom 5 countries for a specific region (Asia). By identifying the "Bottom 5" countries within a high-risk region like Asia, you find real-world models of success that are likely more culturally and economically relevant to their neighbors than Western models. This solution can be used for every country and region.

Conclusion

This report demonstrates a drill down approach to data visualization with story telling and we see how the story unfolds the more we dive deeper into the data with additional filters and variables which is essential for solving complex global health problems.

Our analysis confirms that while BMI and Diabetes trends are closely linked globally, the regional reality varies significantly. Oceania emerges as a high-intensity clinical target due to extreme prevalence rates, whereas Asia requires a digital and infrastructural shift to manage the sheer volume of its affected population. Furthermore, the European success story proves that targeted medical intervention can successfully manage hypertension even as obesity rates rise.

To combat these rising trends, I recommend that we implement a more digital approach and more awareness to these health issues that will cover a large area rather than one to one doctor visits in high-volume regions like Asia and Intensive Clinical Interventions in high-prevalence Red Zones like Oceania. By addressing the specific scale of impact for each region and picking out positive attributes and actions from the countries with lowest prevalence rates global health ministers can build more resilient systems that account for both the percentage of disease and the total number of lives at risk.

Appendix

1. Gender Gap Calculation

This measure was created to identify disparities between genders across different regions.

- **Formula:**

Gender Gap = [Female Prevalence] - [Male Prevalence]

- **Justification:** A positive result indicates higher prevalence in females, while a negative result indicates higher prevalence in males. This measure was used in clustered bar charts to accurately compare categories side-by-side.

2. Population-Weighted Impact (Scale of Impact)

To resolve the "Population Paradox," these formulas were used to calculate the absolute volume of the affected population.

- **Formula Example (Diabetes Impact):**

```
Impact (Regional Scale) =  
VAR_RegionalAvgPrevalence = AVERAGE('Diabetes'[Age-standardised diabetes  
prevalence])  
VAR_RegionalPopulation = SUM('Population'[Population])  
RETURN
```

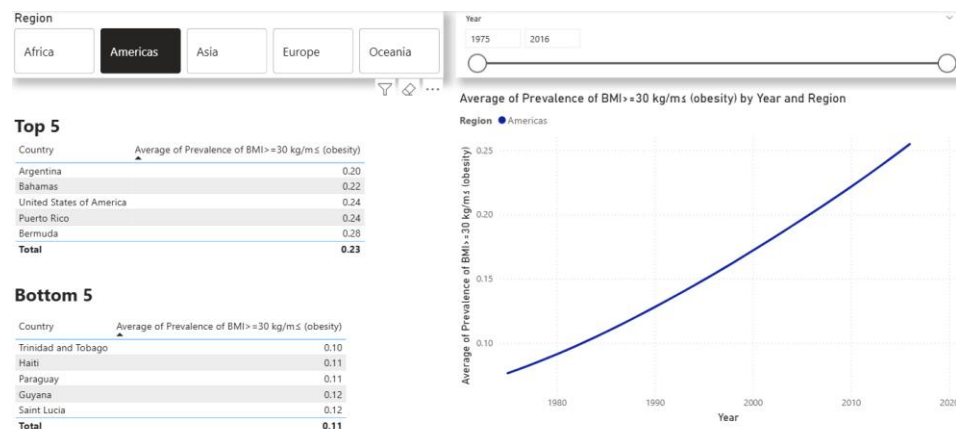
$\text{_RegionalAvgPrevalence} * \text{_RegionalPopulation}$

- **Justification:** Due to data being on a regional level I had to use this formula to calculate the scale of impact this might not be as accurate as having country level population for each country, but this sums up the important insights that we need.

3. Top 5 Dynamic Country Filter (TopN Logic)

4. Filters used for Regions, Year, Sex

Sample Dashboard View with Filters:



References

The following research articles and data sources provided the evidence base for the hypotheses and findings in this report:

- **Article 1 (Clinical Correlation):** Tseng, C. H. (2007). *BMI and hypertension in Taiwanese patients with type 2 diabetes*. PubMed. [Available at: https://pubmed.ncbi.nlm.nih.gov/17965496/](https://pubmed.ncbi.nlm.nih.gov/17965496/).
- **Article 2 (Obesity and Diabetes Link):** Samuel et al. (2022) *Relationship between BMI and Diabetes Prevalence*. PubMed Central (PMC). [Available at: https://pmc.ncbi.nlm.nih.gov/articles/PMC8740746/](https://pmc.ncbi.nlm.nih.gov/articles/PMC8740746/).
- **Data Source (Geographic Hierarchy):** *Country-Region Mapping Table*. FAO / Impact Global Health. [Available at: https://gfinderdata.impactglobalhealth.org/assets/media/table/Country-region-table.xlsx](https://gfinderdata.impactglobalhealth.org/assets/media/table/Country-region-table.xlsx).